

LAB ASSIGNMENT IV
NATURAL LANGUAGE PROCESSING LAB (PMDS606P)
CLASS ID - VL2025260105170

MAXIMUM MARKS: 10

DUE DATE: 25 SEPTEMBER 2025

Problem Statement:

“Obtain a comparative wordlist in CSV format, and write a program that prints those cognates having an edit distance of at least 3 from each other.”

Task Explanation: Comparative Wordlist and Edit Distance Program

1. What is a comparative wordlist?

A comparative wordlist is usually a table of words from two (or more) languages that are being compared for similarities.

For example:

English	German
Water	wasser
Sun	sonne
Mother	mutter
Night	nacht

2. What are cognates?

Cognates are words in two languages that share a common origin and often look/sound similar.

Examples:

- English 'mother' ↔ German 'mutter'
- English 'sun' ↔ German 'sonne'

3. What is edit distance?

The edit distance (Levenshtein distance) between two words is the minimum number of operations (insert, delete, substitute) needed to transform one word into another.

Examples:

- Distance('cat', 'cut') = 1 (substitute 'a' → 'u')
- Distance('kitten', 'sitting') = 3
- Distance('sun', 'sonne') = 2

4. What does the task want?

You will obtain a CSV wordlist (say, comparing English and German words). You need to write a program that:

1. Reads the CSV file.
2. For each row, compares the words (cognates).
3. Finds pairs whose edit distance is at least 3 (≥ 3).
4. Prints those cognates.

5. Example Input (CSV):

English,German
water,wasser
sun,sonne
mother,mutter
night,nacht

6. Expected Output (when edit distance ≥ 3):

- 'water' vs 'wasser' $\rightarrow$ distance = 2 (not printed)
- 'sun' vs 'sonne' $\rightarrow$ distance = 2 (not printed)
- 'mother' vs 'mutter' $\rightarrow$ distance = 2 (not printed)
- 'night' vs 'nacht' $\rightarrow$ distance = 3 (printed)

Output:

night - nacht (distance = 3)

Assessment Task: Comparative Wordlist & Edit Distance

1. Prepare a **CSV file (wordlist.csv)** containing a comparative wordlist of two categories of words.
 - Examples of categories you may choose:
 - Two languages (e.g., English–Hindi, English–French, English–Tamil, etc.)
Examples:
 - ✓ English, Hindi
 - ✓ water, paani
 - ✓ sun, suraj
 - ✓ mother, maa
 - ✓ night, raat
 - British English vs American English
Examples:
 - ✓ colour, color
 - ✓ favourite, favorite
 - ✓ theatre, theatre
 - ✓ defence, defense
 - Formal vs Informal words
Examples:
 - ✓ purchase, buy
 - ✓ assist, help
 - ✓ depart, leave
 - ✓ inform, tell
 - Synonyms
Examples
 - ✓ happy, joyful
 - ✓ big, large
 - ✓ quick, fast
 - ✓ angry, mad

- Technical terms vs Layman terms

Examples

- ✓ cardiologist, heart doctor
- ✓ automobile, car
- ✓ influenza, flu
- ✓ hydration, drinking water

2. Write a Python program that:

- Reads the CSV file.
- Calculates the **edit distance** between the two words in each row. (Manually)
- Prints only those word pairs whose **edit distance** ≥ 3 .